The **Mobile App** Project Canvas (ver. 1.0)

Connecting the Last Mile — Internet for All Territorians

CONCEPT

A hybrid solution addressing the Northern Territory's digital divide by:

1. *Developing an offline-first learning app to give remote students access to educational resources without continuous internet.*
2. *Deploying community-based mesh networks to provide affordable, reliable internet coverage in underserved areas.*

OBJECTIVES & PURPOSES

- *Reduce the digital access gap in pilot communities by at least 30% within the first year.*
- *Improve student access to curriculum content and telehealth services.*
- *Create a scalable model for other rural/remote regions.*

VALUES

1. *Accessibility – Inclusive design for all users, regardless of connectivity.*
2. *Affordability – Shared infrastructure lowers costs for individuals.*
3. *Community Empowerment – Locally managed and maintained networks*

PERSONAS

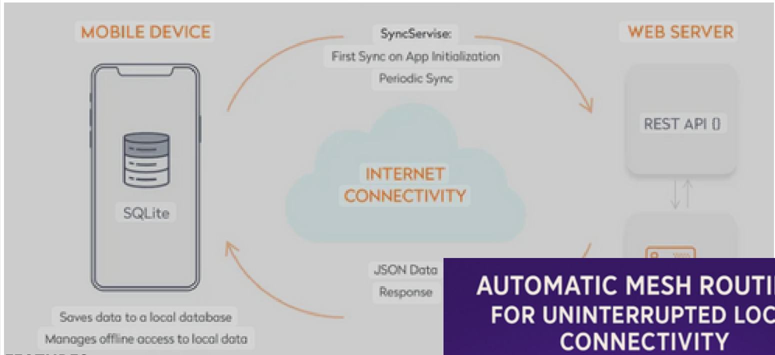
- Remote Students – Children and young adults in rural and Indigenous communities without stable internet access.
- Teachers & Educators – Upload and distribute learning materials that students can use offline.
- Healthcare Workers – Deliver telehealth services with low data usage.
- Community Leaders & Councils – Oversee deployment and sustainability of mesh networks.
- Local IT Volunteers – Assist in setup, troubleshooting, and maintenance.

PROBLEMS TO SOLVE

- **Inconsistent or non-existent internet in remote NT.**
- **High data costs making internet unaffordable.**
- **Education and telehealth access gaps for remote residents.**

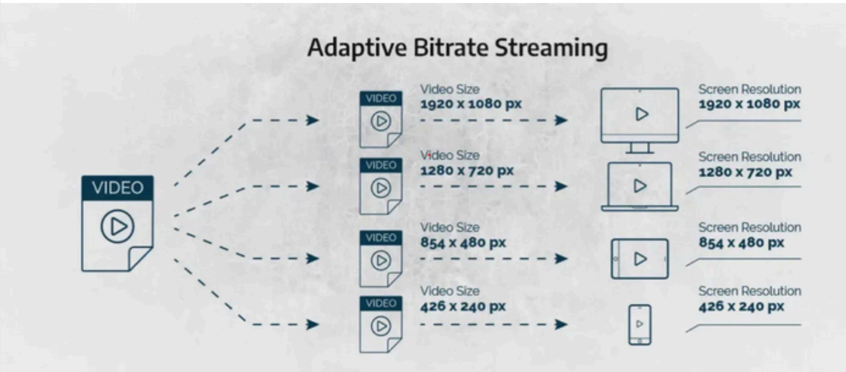
COMPONENTS

1. **Offline Learning App – Educational content, quizzes, video playback, and auto-sync.**
2. **Mesh Network Hardware – Low-cost routers, Raspberry Pi nodes, and satellite uplink.**
3. **Management Dashboard – For network monitoring, bandwidth allocation, and troubleshooting.**



FEATURES

- *Works fully offline with sync-on-connect capability.*
- *Video compression and adaptive streaming to save bandwidth.*
- *Automatic mesh routing for uninterrupted local connectivity.*
- *Shared bandwidth among households to reduce costs.*
- *Easy network setup with plug-and-play nodes.*



MILESTONES

1. *Research & Planning – Identify target community and design network architecture (2 weeks).*
2. *Prototype Learning App – Core offline content and sync function (4 weeks).*
3. *Simulate Mesh Network – Lab-based testing and optimisation (2 weeks).*
4. *Pilot Deployment – Install in one NT community, gather feedback (6 weeks).*
5. *Refinement & Expansion Plan – Incorporate feedback and prepare Code Fair presentation.*

STAKEHOLDERS

- **NT Government – Digital Strategy & Education Departments – Provide funding and policy support.**
- **Community Councils – Approve and manage deployments.**
- **Schools & Training Centres – Integrate the app into the curriculum.**
- **Healthcare Services – Use connectivity for telehealth.**
- **Technology Partners – Supply routers, uplink, and app development support.**

RISKS

- **Difficulty sourcing affordable, durable hardware.**
- **Limited local technical expertise for maintenance.**
- **Weather and environmental challenges for outdoor equipment.**
- **Funding sustainability for long-term operations.**

DELIVERABLES

- **Progressive Web App (PWA) or APK for learning platform.**
- **Mesh network simulation and hardware setup documentation.**
- **Poster and presentation for CDU IT Code Fair 2025.**
- **Community training materials.**

APP NAME

Last Mile

CONTEXT OF USAGE &COVERAGE

- *Offline-first mode for a learning app with local data storage.*
- *Designed for low-bandwidth, high-latency environments common in NT's remote areas.*

TECHNOLOGY

- *Frontend: Flutter or React for PWA.*
- *Backend: Node.js with offline sync API.*
- *Network Simulation: NS-3 or Cisco Packet Tracer.*
- *Hardware: Raspberry Pi, Ubiquiti routers.*

PLATFORM,OS, ...

- *Android smartphones/tablets (primary).*
- *Web browsers (secondary).*

ORIENTATION

Both portrait and landscape support

RELEASE

- *Pilot in selected NT communities via a local council partnership.*
- *Public release as a free app with open-source mesh network guidelines.*

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